Frontier



Bahia Verde Minerals

Carbon dioxide removal prepurchase application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Bahia Verde Minerals

Organization location (we welcome applicants from anywhere in the world)

Bahia, Brazil

Name(s) of primary point(s) of contact for this application

Steve Norcini

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

For-profit company transforming hypersaline brines in Bahia, Brazil, into ultrapure chemicals and durable carbon removal products.

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Brine-Derived Carbonates for Durable Direct Air Mineralization (DAM)

[1] We use “project” throughout this template, but the term is not intended to denote a single facility. The “project” being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your organization.

b. **Description of the CDR approach:** Describe how the proposed technology removes CO_2 from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us how your project addresses any of the priority innovation areas identified in the RFP. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Bahia Verde is developing a brine-derived carbon mineralization platform that removes CO_2 from the atmosphere by converting naturally abundant magnesium and calcium in hypersaline brine into stable carbonate minerals. The project uses selective brine refining, controlled crystallization, base-mediated mineral recovery, and engineered air-contact carbonation to produce durable Mg-carbonates and $CaCO_3$ products. The carbon removal claim is based only on atmospheric CO_2 that is converted into solid inorganic carbonate minerals and placed into long-lived products or engineered mineral storage. Co-products and avoided emissions may improve project economics, but they will not be counted as CDR.

The core CDR mechanism is mineral carbonation. In the magnesium pathway, magnesium is first recovered from brine as a high-purity alkaline solid, primarily magnesium hydroxide, $Mg(OH)_2$. This material is then exposed to atmospheric CO_2 in engineered air-contacting systems. Under controlled moisture, particle-size, and residence-time conditions, CO_2 reacts with $Mg(OH)_2$ to form $MgCO_3$ and hydrated magnesium carbonate phases. A simplified reaction is:

$$\mathbf{Mg(OH)_2 + CO_2 \rightarrow MgCO_3 + H_2O}$$

Depending on humidity, exposure time, and product morphology, the final magnesium product may include hydrated Mg-carbonates such as nesquehonite-, dypingite-, or hydromagnesite-type phases. These are solid inorganic carbonate minerals and can store carbon for more than 1,000 years when placed into durable applications or controlled mineral storage.

In the calcium pathway, calcium-rich brine is treated through base-assisted air-contact carbonation to precipitate calcium carbonate, $CaCO_3$. Atmospheric CO_2 is introduced through air contact, and the calcium-bearing solution is converted into solid $CaCO_3$ products. A simplified representation is:

$$\mathbf{Ca^{2+} + CO_2 + alkaline\ conditions \rightarrow CaCO_3(s)}$$

The $CaCO_3$ product can be produced as precipitated calcium carbonate, carbonate aggregate, filler, panel ingredient, composite additive, or engineered mineral-storage material. Because $CaCO_3$ is a stable mineral form of carbon, it provides a second durable carbon-storage route parallel to the magnesium pathway.

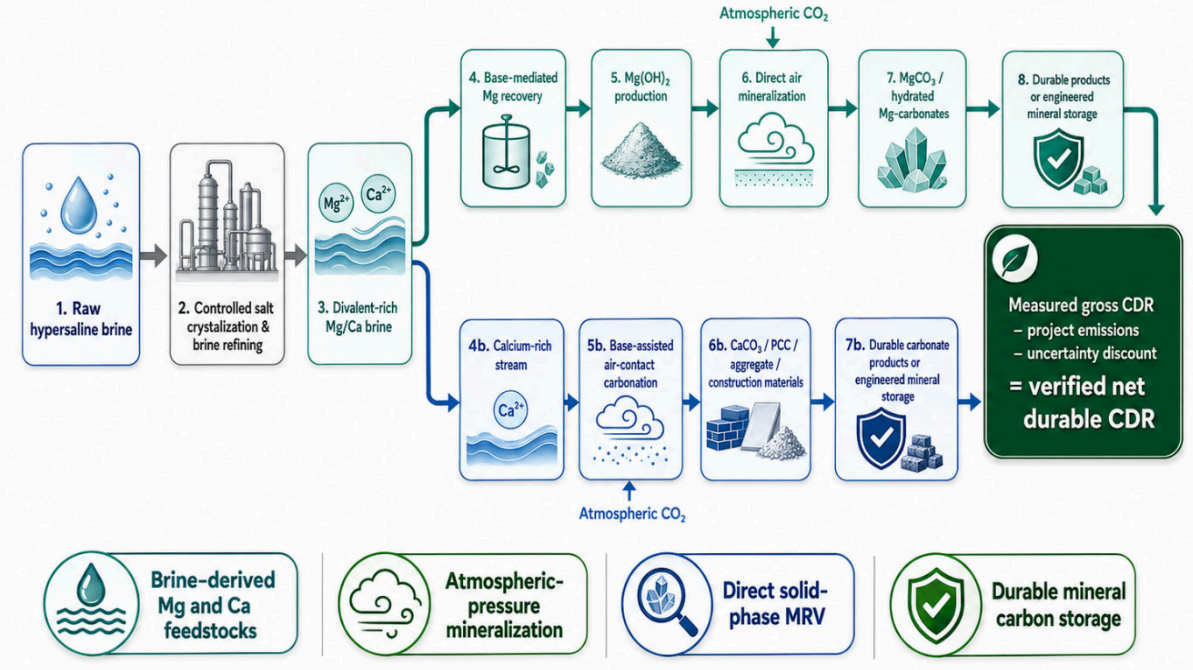
The process begins with raw hypersaline brine containing sodium, potassium, magnesium, calcium, chloride, and trace elements. Rather than treating this brine as waste, Bahia Verde uses it as a concentrated mineral feedstock. The first step is controlled salt crystallization, which selectively removes a portion of monovalent salts as saleable co-products and simplifies the remaining brine chemistry. This produces a divalent-rich brine enriched in magnesium and calcium. Removing monovalent salts early improves downstream selectivity, reduces ionic complexity, and supports higher-purity mineral products.

The divalent-rich brine is then treated with a low-carbon base to selectively recover magnesium as $Mg(OH)_2$. The $Mg(OH)_2$ solid is separated, washed, and directed to the magnesium carbon-mineralization pathway. This step is important because $Mg(OH)_2$ contains reactive alkalinity in

a solid form. Unlike raw silicate minerals that may require slow weathering or intensive comminution, $Mg(OH)_2$ can react directly with atmospheric CO_2 under mild conditions. Bahia Verde has observed rapid carbonation behavior under atmospheric pressure and ambient operating conditions, without requiring high-pressure reactors, high-temperature treatment, or concentrated CO_2 as the only carbon source.

After magnesium recovery, the calcium-rich brine is processed through an air-contact carbonation step. A low-carbon base is used to control pH and drive calcium carbonate precipitation while air supplies the CO_2 source. The calcium pathway is especially important because the brine contains a high calcium concentration, creating a large theoretical capacity for durable carbonate formation. Product characteristics such as particle size, morphology, purity, and end-use suitability can be tuned by controlling pH, base addition, CO_2 contact rate, mixing, residence time, washing, and drying (Figure 1).

Figure 1. Public process schematic for brine-derived atmospheric CO_2 mineralization



This project addresses several of Frontier's priority innovation areas. First, it advances surficial mineralization by producing reactive alkaline materials from brine rather than relying on naturally slow weathering of raw rock. Second, it develops direct air-contacting mineralization systems for Mg- and Ca-bearing materials. Atmospheric CO_2 is converted directly into solid carbonates without first producing a purified, compressed CO_2 stream. Third, it demonstrates a brine-derived hydroxide and carbonate feedstock strategy, using salt-rich resources as a source of alkalinity for CO_2 mineralization. This is highly aligned with the need for scalable, low-cost mineralization pathways that avoid dependence on mined ultramafic rock, long-distance transport of feedstock, or disposal of low-value solids.

Bahia Verde is differentiated from conventional direct air capture because the process does not

require cyclic sorbent regeneration, high-temperature desorption, vacuum swing, high-pressure CO_2 compression, pipeline transport, or geologic injection. The capture medium and storage medium are integrated: atmospheric CO_2 reacts with brine-derived alkaline materials and becomes part of a solid carbonate product. This reduces the number of unit operations and creates a physically inspectable storage product that can be weighed, sampled, chemically analyzed, and tracked.

The system is also differentiated from enhanced rock weathering. Enhanced weathering often depends on spreading ground silicate minerals into open environments and inferring removal from soil, water, or drainage chemistry over long periods. Bahia Verde's approach is an engineered ex-situ process. The feedstock is refined before carbonation, reaction conditions are controlled, and the carbon is measured directly in recovered solid products. This reduces uncertainty from hydrology, soil variability, secondary mineral formation, and dispersed product fate.

A further differentiation is product value. Many mineralization approaches generate carbonate solids that must be stored at low value. Bahia Verde is designed to produce durable, saleable carbonate products. Calcium carbonate can be used as PCC, aggregate, high quality construction material, filler, panel ingredient, or composite additive. Magnesium carbonate and hydrated Mg-carbonates can be used in premium construction materials, specialty mineral fillers, panels, composites, or engineered storage. These product pathways create the potential to reduce net CDR cost while maintaining conservative carbon accounting. Co-products from the brine-refining process may also support project economics, but they will be reported separately and will not be counted as carbon removal.

Durability is based on conversion of atmospheric CO_2 into solid inorganic carbonate minerals. The project will only credit carbon stored as Mg-carbonates or $CaCO_3$ in durable products or controlled mineral storage. Carbon associated with intermediate soluble species, short-lived uses, avoided emissions, or co-product substitution will not be counted as CDR. Product pathways that could release CO_2 through calcination, acid neutralization, or intentional decomposition will be excluded from credited CDR.

Measurement, reporting, and verification will be based on batch-level solid-carbon accounting. Gross CDR will be calculated from measured inorganic carbon in final carbonate products relative to starting materials. Each credited batch will be characterized by product mass, moisture content, total inorganic carbon, Ca/Mg composition, carbonate mineralogy, and product destination. Analytical tools will include inorganic carbon analysis, XRD, XRF, Raman, elemental analysis, and mass balance. Net CDR will subtract cradle-to-grave emissions from brine handling, electricity, base production, pumping, mixing, separation, air movement, drying, transport, product finishing, and MRV. A conservative uncertainty discount will then be applied.

In summary, Bahia Verde removes atmospheric CO_2 by converting brine-derived magnesium and calcium into durable carbonate minerals. Its advantages are high-reactivity brine-derived feedstocks, mild-condition mineralization, direct solid-phase MRV, valuable carbonate products, and a modular process that can scale with brine production. This creates a differentiated pathway for durable, measurable, and potentially low-cost carbon dioxide removal.

c. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale

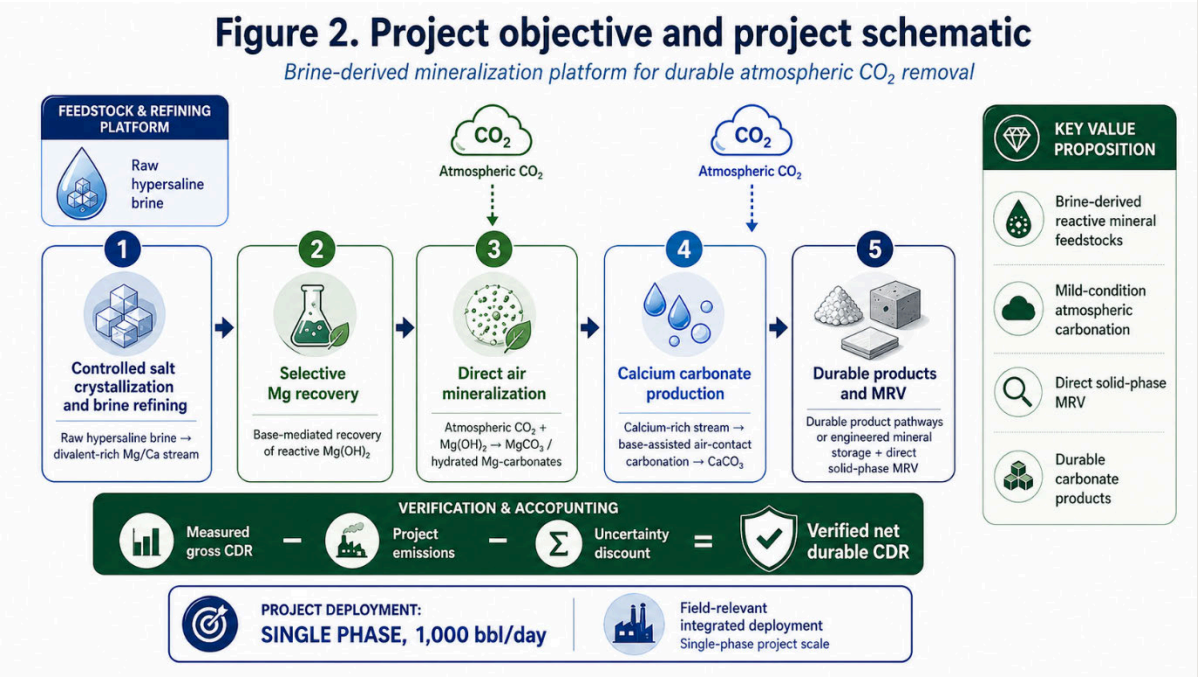
criteria?[2] What is your approach to quantifying the carbon removed? Please include figures and system schematics, and be specific but concise. 1000-1500 words

Bahia Verde is building a modular brine-derived carbon mineralization platform that converts hypersaline brine into durable magnesium and calcium carbonate products while generating saleable co-products from the same brine-refining system. The objective of the Frontier project is to demonstrate that naturally concentrated magnesium and calcium in brine can be transformed into high-purity reactive mineral feedstocks, contacted with atmospheric CO_2 under mild conditions, and converted into solid carbonate minerals that provide durable carbon dioxide removal.

The project is designed to move from an already demonstrated brine-refining platform toward a field-relevant CDR system. The core brine-refining and mineral-production technology has operated at approximately 1 m^3/hour and produced high-purity mineral products with minimum purity of 99.9%. The proposed Frontier project will integrate this platform with atmospheric mineralization, product qualification, cradle-to-grave LCA, and batch-level MRV. The first objective is not only to maximize early tonnage, but to generate a high-confidence operating and verification package showing that brine-derived Mg and Ca streams can be converted into durable carbonate products with auditable net CDR.

The system being built has five major elements. First, raw hypersaline brine is processed through controlled salt crystallization and brine refining to reduce ionic complexity and produce a divalent-rich Mg/Ca stream. Second, magnesium is selectively recovered as a reactive alkaline material, primarily $Mg(OH)_2$. Third, $Mg(OH)_2$ is exposed to atmospheric CO_2 in engineered direct-air mineralization systems to form $MgCO_3$ and hydrated Mg-carbonates. Fourth, calcium-rich streams are converted into $CaCO_3$ through base-assisted air-contact carbonation. Fifth, the Mg- and Ca-carbonate products are directed to durable product markets or engineered mineral storage, while MRV quantifies the carbon stored in solid products (Figure 2).

[2] We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at $100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.



The initial deployment will be associated with the Bahia Verde brine resource and will use a design basis of 1,000 bbl/day, equivalent to approximately 159 m^3/day of brine. At this scale, the final TEA estimates a gross CDR capacity of approximately 4,347 tCO_2/year from the Ca- and Mg-carbonate pathways. After subtracting estimated annual project emissions of 960 tCO_2e/year, the project yields approximately 3,387 tCO_2/year of pre-discount net removal. Applying a conservative 10% uncertainty discount gives approximately 3,049 tCO_2/year of credited net CDR.

The system is designed to scale through a modular multi-well architecture. At a representative commercial configuration of approximately 4,000 m^3/day per well, modeled gross CDR scales to approximately 109,000 tCO_2/year per well, with approximately 76,700 tCO_2/year of credited net CDR under the same process assumptions. While the Bahia Verde resource alone does not reach Frontier's full 0.5 Gt/year target, the broader opportunity is not limited to a single well or reservoir.

Calcium- and magnesium-rich brines are widely distributed across global sedimentary basins, particularly in oil and gas systems, evaporitic formations, and deep saline aquifers. Produced water from hydrocarbon operations alone exceeds hundreds of millions of cubic meters per day globally, and many of these streams contain significant concentrations of Ca^{2+} and Mg^{2+}. In addition, large subsurface hypersaline systems comparable to the Tucano Basin exist in regions such as the Permian Basin, the Smackover Formation, parts of South America, the Middle East, and Europe. These brines represent naturally concentrated, continuously replenished fluid systems that can serve as feedstocks for mineral-based carbon removal.

The current cost structure is typical of a first-of-a-kind integrated mineralization project. The main cost categories are capital equipment for brine refining, controlled crystallization, solids handling, air-contact systems, and MRV instrumentation; fixed OpEx for labor, maintenance, and site operations; energy for pumping, mixing, separation, air movement, and product finishing; and variable OpEx for low-carbon base/reagent input, consumables, filtration, product handling, logistics, and process make-up. The final TEA estimates a levelized cost of approximately \$1,741/tCO_2 for this project. The

cost breakdown is approximately \$444/tCO_2 CapEx, \$335/tCO_2 fixed OpEx, \$14/tCO_2 energy, and \$978/tCO_2 other variable OpEx. CO_2 transport and storage costs are zero because the project stores carbon directly in solid carbonate products rather than in a compressed CO_2 stream.

The dominant cost driver is other variable OpEx, particularly low-carbon base/reagent demand and associated product-processing costs. Therefore, the path to lower cost depends primarily on reagent efficiency, recovery/recycle, continuous operation, improved crystallization efficiency, high Mg/Ca recovery, faster atmospheric carbonation, reduced drying and finishing cost, and offtake-backed sale of durable carbonate products and co-products. The final TEA estimates a FOAK TRL-9 cost of approximately \$1,397/tCO_2, showing meaningful reduction from the first project but also indicating that learning-rate-driven CapEx reduction alone will not reach Frontier's long-term cost target.

For the process to approach Frontier's long-term cost and scale criteria, Bahia Verde must operate as an integrated brine-refining and CDR platform, not as a stand-alone carbon-storage process. The major cost-down levers are continuous operation, larger modular trains, improved base efficiency, higher product recovery, lower reagent loss, low-cost renewable power, optimized air-contact systems, lower MRV cost through validated sampling plans, and co-product revenue. Product qualification is therefore a core project objective. Mg- and Ca-carbonate products may enter durable applications such as PCC, aggregate, construction materials, fillers, panels, composites, and engineered mineral storage. Avoided emissions and co-product substitution benefits will be reported separately and will not be counted as CDR.

Quantification of carbon removed will use direct solid-phase MRV. Gross CDR will be calculated from measured inorganic carbon in final carbonate products relative to starting materials. Each credited batch will be tracked by brine input, product mass, moisture content, total inorganic carbon, Mg/Ca composition, carbonate mineralogy, and product destination. Analytical tools will include total inorganic carbon analysis, XRD, XRF, Raman, FTIR, TGA, elemental analysis, and mass balance. The project will not rely only on theoretical stoichiometry. Stoichiometric capacity is used for design and planning, but CDR credit will be based on measured carbon in durable solid carbonate products.

Net CDR will be calculated as:

```
Net CDR = measured gross carbonate carbon stored - cradle-to-grave project emissions - uncertainty discount
```

The LCA boundary will include brine handling, electricity, low-carbon base/reagent production, pumping, mixing, separation, air movement, product washing, drying, finishing, transport, equipment allocation, and MRV. Carbon in intermediate or non-durable forms will not be credited unless it is converted into durable Mg- or Ca-carbonates and its product fate is controlled.

In summary, Bahia Verde is trying to build a replicable brine-to-carbonate CDR platform that combines mineral recovery, atmospheric-pressure mineralization, durable carbonate products, and direct solid-phase MRV. The first Frontier project will validate the integrated system at 1,000 bbl/day, with annual credited net CDR capacity of approximately 3,049 tCO_2/year after project emissions and uncertainty discount. Future deployments are designed to reduce cost through scale, modular replication, improved reagent efficiency, higher uptime, product qualification, and co-product-backed economics.

d. **Risks:** What are the biggest risks and how will you mitigate those? Include technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. 500-1000 words

Bahia Verde's proposed project has several important risks, but the single-phase 1,000 bbl/day deployment is intentionally sized to reduce these risks while generating the technical, MRV, product, and cost data needed for future scale-up. The project's risk strategy is to avoid overclaiming early tons, rely on direct solid-phase carbon measurement, maintain conservative LCA boundaries, and preserve flexibility in product pathways.

The first technical risk is brine variability and separation performance. Hypersaline brines can vary in salinity, Mg/Ca ratio, trace elements, suspended solids, and scaling tendency. These variations could affect mineral recovery, product purity, equipment fouling, and reagent demand. Bahia Verde will mitigate this by implementing routine brine characterization, front-end pretreatment as needed, operating envelopes for density, pH, and conductivity, and staged quality-control checks after each major separation step. The existing brine-refining platform has already demonstrated production of high-purity mineral products, and the project will use the 1,000 bbl/day deployment to validate continuous performance with real field brine.

A second technical risk is carbonate reaction kinetics and conversion efficiency. Although rapid carbonation has been observed under mild, atmospheric-pressure conditions, full-scale conversion depends on particle size, humidity, air-contact design, residence time, mixing, and product morphology. Incomplete carbonation would reduce gross CDR and increase cost per ton. This will be mitigated through controlled air-contact testing, optimization of solids geometry, humidity and exposure control, recycling of under-carbonated material, and batch-level verification of actual carbonate formation. The project will claim only measured carbonate carbon, not theoretical capacity.

A third technical risk is product quality and end-use acceptance. The platform depends on directing Mg- and Ca-carbonate products into durable applications such as PCC, aggregate, construction materials, fillers, panels, composites, or engineered mineral storage. These markets may require specific purity, particle size, moisture, whiteness, chloride content, leaching behavior, or mechanical performance. Bahia Verde will mitigate this through product qualification, customer/offtake engagement, washing and finishing controls, and fallback pathways into engineered mineral storage if premium product markets are slower to develop.

The main MRV risk is overstating durable carbon removal. The process may produce multiple products and intermediate streams, but only carbon stored in durable Mg- and Ca-carbonate solids will be credited. No credit will be taken for avoided emissions, product substitution, non-durable intermediates, or co-products. Gross CDR will be measured using product mass, moisture correction, total inorganic carbon, Ca/Mg elemental balance, XRD, FTIR, TGA, and batch records. Net CDR will subtract cradle-to-grave emissions and apply a conservative uncertainty discount. This approach directly mitigates double counting, analytical uncertainty, incomplete conversion, and product-fate uncertainty.

A related durability risk is reversal through unsuitable product use. Carbonate products could release CO_2 if sent to high-temperature calcination, acid-neutralization markets, or other decomposition pathways. Bahia Verde will exclude such end uses from credited CDR, use product-use restrictions where applicable, and track credited batches to durable products or engineered mineral storage. The durability claim will be tied to the mineral form and final product pathway, not simply to production

volume.

Project execution risks include equipment procurement, integration of unit operations, commissioning delays, solids handling, and operational reliability. The process includes brine refining, base-mediated mineral recovery, air-contact mineralization, product finishing, and MRV. Integration risk is managed by using a single field-relevant deployment scale rather than attempting immediate commercial expansion. Critical milestones will include final engineering, vendor data, product specifications, commissioning tests, and early verified batches before full delivery.

Ecosystem and health risks are expected to be manageable but require careful controls. Potential risks include brine spills, high-salinity residues, dust from dry mineral products, trace-element concentration, and leaching from products. Mitigation measures include secondary containment, closed or controlled brine handling, dust suppression, worker exposure controls, product leach testing, trace-element monitoring, and appropriate storage/product-use criteria. Environmental impacts will be evaluated within the LCA and operational monitoring plan.

Financial risk is driven by CapEx, reagent/base cost, energy use, product finishing, and co-product revenue realization. The TEA indicates first-project costs remain above long-term targets, so cost reduction depends on process optimization, reagent efficiency, higher uptime, and durable product offtake. Bahia Verde will mitigate financial risk through conservative TEA assumptions, staged procurement, product qualification, multiple revenue pathways, and separating CDR revenue from co-product revenue. The first deployment is therefore structured to be technically rigorous, commercially informative, and appropriately conservative for Frontier prepurchase delivery.

e. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) *(Should be net volume after taking into account the uncertainty discount proposed in 5c)*	3,049 net tCO_2 after applying the proposed 10% uncertainty discount. This is based on the single-phase 1,000 bbl/day deployment.
Delivery window *(At what point should Frontier consider your contract complete? Should match 2f)*	Q3 2027–Q4 2028. Frontier should consider the contract complete after verified delivery of 3,049 net tCO_2 from durable Mg- and Ca-carbonate products, after project emissions and uncertainty discount.
Levelized cost ($/ton CO_2) *(This is the cost per ton for the project tonnage described above, and should match 6d)*	$1,741/$tCO_2$, based on the formula-driven TEA for the 1,000 bbl/day project.

Levelized price ($/ton CO_2)[3] *(This is the price per ton of your offer to us for the tonnage described above)*	$1,400/t$CO_2$ offered to Frontier. This price reflects the early-stage FOAK nature of the project while accounting for anticipated co-product value and the strategic value of Frontier support.

[3] This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).